CSS 311 Parallel and Distributed Computing

ASSIGNMENT-2

1. It is the **extension** of Assignment-1.
2. Module-III to be completed for Assignment-2.
3. Apply **Foster's Methodology** for Parallel algorithm design with necessary steps. **(Diagram Illustration is must with proper explanation)**
4. MPI Implementation and Solution demonstration.

MODULE-I [Completed in Assignment-I]

1. **Objective and Description**
   a) Articulate the objective of the problem.
   b) Provide a comprehensive description answering key questions such as what, why, and where.
2. **Pseudocode/Algorithm/Flowchart**

   Develop the pseudocode, algorithm, or flowchart for the identified problem.
3. **Solution Demonstration**

   Demonstrate the solution for the chosen problem using small test cases to illustrate its functionality.
4. **Serial Code Implementation (C/C++)**
   a) Implement the serial code for the problem using C/C++.
   b) Display the required output, incorporating a minimum of three test cases and considering all aspects of the problem. Increase the dimension of the input size and report the results for the same.
5. **Time Analysis**
   a) Derive time analysis for the implemented program. Asymptotic complexity analysis for different test cases.

MODULE-II [Completed in Assignment-I]

Parallel Version - OpenMP

MODULE-III [Repeat Module-I, then do this Module]

Parallel Version - MPI

6. **Identification of Parallelizable Blocks** [Apply Foster's Design Methodology]

   Determine the segments of the chosen problem that can be parallelized.

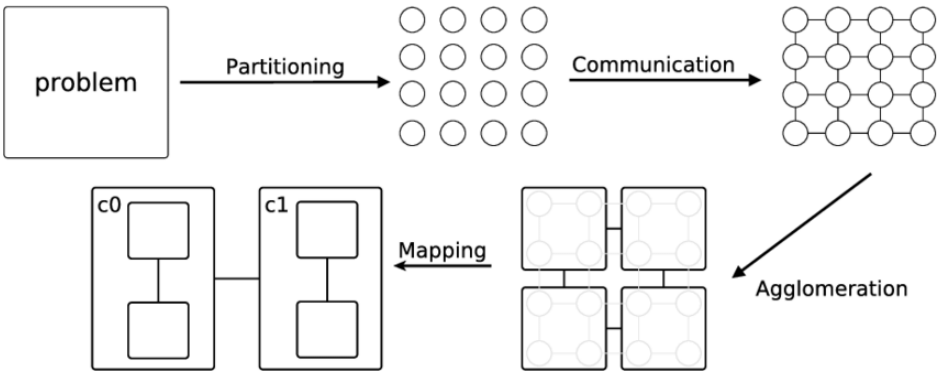


Fig. 1: Foster's Design Methodology

7. **Pseudocode for Parallel Version with respect to MPI**

   Develop the pseudocode outlining the parallelized version of the solution.

8. **Solution Demonstration with respect to MPI**

   Demonstrate the solution for the chosen problem using small test cases to illustrate its functionality.

9. **MPI Parallel Code Implementation (C/C++)**

   a) Implement the parallel code using OpenMP in C/C++.

   b) Display the output, incorporating a minimum of three test cases and considering all aspects of the problem. Increase the dimension of the input size and report the results for the same.

10. **Time Analysis for MPI Implementation**

    Conduct asymptotic complexity analysis for different test cases.

11. **Execution Time Comparison Table and Graph**

    Generate a Table and graph comparing the execution times of the serial code, parallel code using OpenMP, and parallel code using MPI.

Instructions

1. Review 1 - 15/10/2026 [MODULE-I and MODULE-III]

2. Final Online Submission in LMS - 19/10/2026, 5:30PM

3. **Presentation** as Assignment-1 will starts from 19/10/2026. The schedule will be shuffled for this presentation. Prepare the **PPT** in advance.

4. You may answer the questions 1, 2, 3, 5 from MODULE-I and 6, 7, 8, 10 from MODULE-III using paper and pen **[A4 Sheet]**. [Clear and understandable **handwritten**]

5. Create a **Single PDF** and upload it in **LMS** and submit the **hardcopy** of the original while presentation. It contains the following 4 files:
    i. **Handwritten** Answers with questions. [1, 2, 3, 5 from **MODULE-I** and 6, 7, 8, 10 from **MODULE-III**.
    ii. Serial Code with output (Name and Roll No) should be displayed. [**Attach** in the appropriate place in Hardcopy]
    iii. Parallel Code-MPI with output (Name and Roll No) should be displayed. [**Attach** in the appropriate place in Hardcopy]

6. If the above procedure is **not followed** then the submitted assignment cannot be evaluated and **no mark** is awarded for the same.

**********ALL THE BEST*********